

Belief State Updates for a Switching Mess3 Process

Corrected 18-Dimensional Formulation

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1 Single-Process Notation (Riechers et al.)

A **Mess3 process** P_i has hidden states $\sigma \in \{0, 1, 2\}$, token alphabet $\mathcal{X} = \{A, B, C\}$, transition matrix $T^{(i)}$, and emission probabilities $p_i(x | \sigma)$.

Definition 1.1 (Symbol-labelled transition matrix). For token $x \in \mathcal{X}$, the symbol-labelled transition matrix of process P_i is

$$T_x^{(i)} = T^{(i)} \cdot \text{diag}(p_i(x | 0), p_i(x | 1), p_i(x | 2)) \in \mathbb{R}^{3 \times 3}.$$

Entry $T_x^{(i)}[\sigma, \sigma']$ is the probability of transitioning from σ to σ' and emitting x .

Definition 1.2 (Single-process belief update). The belief state is a row vector $\boldsymbol{\eta}_t \in \Delta_2$. Upon observing token x ,

$$\boldsymbol{\eta}_{t+1} = \frac{\boldsymbol{\eta}_t T_x^{(i)}}{Z}, \quad Z = \boldsymbol{\eta}_t T_x^{(i)} \mathbf{1}.$$

2 Setup: Two Processes and Regime Switching

We study two Mess3 processes $P_1 \sim \text{Mess3}(\alpha_1, \chi_1)$ and $P_2 \sim \text{Mess3}(\alpha_2, \chi_2)$ with a regime variable $r_t \in \{1, 2\}$ indicating which process is active (emitting) at time t .

Variant [2a] — frozen inactive state. When process P_j is inactive its hidden Markov chain is frozen; its state is preserved exactly from the last time it was active and resumes from that saved state when next activated.

Switching. Time is divided into blocks of length m . At the end of each block the regime switches according to the 2×2 stochastic matrix

$$R = \begin{pmatrix} 1-p & p \\ p & 1-p \end{pmatrix},$$

where $p \in [0, 1]$ is the switching probability (symmetric here for simplicity). The stationary distribution of R is $\boldsymbol{\pi}_R = [1/2, 1/2]$.

3 The 18-Dimensional Belief State

3.1 Full hidden state

The full hidden state at time t is the triple $(r_t, \sigma_t^{(1)}, \sigma_t^{(2)})$, where $r_t \in \{1, 2\}$ is the active regime and $\sigma_t^{(i)} \in \{0, 1, 2\}$ is the hidden Markov state of process P_i (frozen when P_i is inactive). The joint state space has $2 \times 3 \times 3 = 18$ elements.

3.2 Belief state as a ket

We write the belief state as a ket in $\mathbb{R}^{18}$ using the tensor product structure $\mathbb{R}^2 \otimes \mathbb{R}^3 \otimes \mathbb{R}^3$:

Definition 3.1 (Belief ket).

$$\langle \psi(t) | \in \mathbb{R}^2 \otimes \mathbb{R}^3 \otimes \mathbb{R}^3,$$

with entries

$$\langle \psi(t) | [r, s_1, s_2] = P(r_t = r, \sigma_t^{(1)} = s_1, \sigma_t^{(2)} = s_2 \mid x_{1:t}).$$

The three tensor factors correspond to: (i) regime, (ii) state of P_1 , (iii) state of P_2 . All entries are non-negative and sum to one.

3.3 Initial state

At $t = 0$, the regime is unknown (equal prior) and each process is at its own stationary distribution $\pi^{(i)}$ (the left eigenvector of $T^{(i)}$ with eigenvalue 1):

$$\langle \psi(0) | = \langle r(0) | \otimes \langle \eta_0^{(1)} | \otimes \langle \eta_0^{(2)} |, \quad \langle r(0) | = [\tfrac{1}{2} \quad \tfrac{1}{2}], \quad \langle \eta_0^{(i)} | = \pi^{(i)}. \quad (1)$$

This is a product state. After the first observation with regime uncertainty, the belief is generically *entangled* (non-product) in the regime-chain tensor product.

4 Within-Block Update

4.1 Update operator

Inside a block the regime does not switch. When token x is observed, only the *active* chain transitions; the inactive chain is frozen (identity). In the $\mathbb{R}^2 \otimes \mathbb{R}^3 \otimes \mathbb{R}^3$ tensor product, the per-regime projectors $|1\rangle\langle 1|$ and $|2\rangle\langle 2|$ select which chain is active:

Unnormalised within-block update operator for token x :

$$U(x) = \underbrace{|1\rangle\langle 1| \otimes T_x^{(1)} \otimes I_3}_{\text{regime 1 active: } P_1 \text{ emits, } P_2 \text{ frozen}} + \underbrace{|2\rangle\langle 2| \otimes I_3 \otimes T_x^{(2)}}_{\text{regime 2 active: } P_2 \text{ emits, } P_1 \text{ frozen}}. \quad (2)$$

$U(x)$ is an 18×18 block-diagonal matrix (in the regime index). Entry $U(x)[r, s_1, s_2; r', s'_1, s'_2]$ is nonzero only when $r = r'$, reflecting the fact that the regime does not change within a block.

4.2 Per-regime likelihoods

Define the marginal token likelihood under each regime:

$$Z_1 = \langle \eta_t^{(1)} | T_x^{(1)} \mathbf{1} = P(x \mid r_t = 1, x_{1:t}), \quad (3)$$

$$Z_2 = \langle \eta_t^{(2)} | T_x^{(2)} \mathbf{1} = P(x \mid r_t = 2, x_{1:t}), \quad (4)$$

where $\langle \eta_t^{(i)} |$ is the marginal belief over $\sigma^{(i)}$ conditioned on regime i being active.

4.3 Global normaliser

The global normaliser is the *regime-weighted* sum:

$$Z = \lambda_t Z_1 + (1 - \lambda_t) Z_2 = P(x_{t+1} \mid x_{1:t}), \quad (5)$$

where $\lambda_t = P(r_t = 1 \mid x_{1:t})$ is the current regime weight.

Critical error to avoid. Normalising each regime term separately — dividing the $|1\rangle\langle 1|$ term by Z_1 and the $|2\rangle\langle 2|$ term by Z_2 independently — is *wrong*. It forces $\lambda_t = 1/2$ permanently and destroys all Bayesian inference about the active regime. The single global Z in (5) is mandatory.

4.4 Normalised within-block update

Corrected within-block update:

$$\langle\psi(t+1)| = \langle\psi(t)| \frac{U(x_{t+1})}{Z}, \quad (6)$$

where $U(x)$ is given by (2) and Z by (5). After this update, the new regime weight is

$$\lambda_{t+1} = \frac{\lambda_t Z_1}{Z}. \quad (7)$$

4.5 State after the first token

Starting from the product initial state (1):

$$\langle\psi(1)| = \frac{1}{Z} \langle\psi(0)| U(x_1) = \frac{1}{Z} \left(\frac{1}{2} \langle 1| \otimes \langle\eta_0^{(1)}| T_{x_1}^{(1)} \otimes \langle\eta_0^{(2)}| + \frac{1}{2} \langle 2| \otimes \langle\eta_0^{(1)}| \otimes \langle\eta_0^{(2)}| T_{x_1}^{(2)} \right), \quad (8)$$

where $Z = \frac{1}{2}Z_1 + \frac{1}{2}Z_2$. This is no longer a product state: the P_2 frozen state is correlated with the P_1 active state via the normaliser Z .

4.6 State after $m - 1$ within-block tokens

Applying (6) repeatedly for tokens $x_1, \dots, x_{m-1}$ (still within the first block, no switch yet):

$$\langle\psi(m-1)| = \frac{1}{Z_{1:m-1}} \left[\frac{1}{2} \langle 1| \otimes \langle\eta_0^{(1)}| T_{x_1}^{(1)} \dots T_{x_{m-1}}^{(1)} \otimes \langle\eta_0^{(2)}| + \frac{1}{2} \langle 2| \otimes \langle\eta_0^{(1)}| \otimes \langle\eta_0^{(2)}| T_{x_1}^{(2)} \dots T_{x_{m-1}}^{(2)} \right], \quad (9)$$

where the global normaliser over the block is

$$Z_{1:m-1} = \frac{1}{2} \langle\eta_0^{(1)}| T_{x_1}^{(1)} \dots T_{x_{m-1}}^{(1)} \mathbf{1} + \frac{1}{2} \langle\eta_0^{(2)}| T_{x_1}^{(2)} \dots T_{x_{m-1}}^{(2)} \mathbf{1}.$$

5 Switch Boundary Update

5.1 Switch operator

At the end of each block of length m , the regime transitions according to R while both chain states are carried forward unchanged. In the tensor product structure this is:

$$S = R \otimes I_3 \otimes I_3 = R \otimes I_9. \quad (10)$$

Explicitly:

$$S = \begin{pmatrix} (1-p)I_9 & pI_9 \\ pI_9 & (1-p)I_9 \end{pmatrix}. \quad (11)$$

Switch boundary update (no normalisation needed):

$$\langle \psi_{\text{after}} | = \langle \psi_{\text{before}} | S = \langle \psi_{\text{before}} | (R \otimes I_9). \quad (12)$$

S is a stochastic matrix, so total probability is preserved exactly. No normalisation step is required.

5.2 Four-path expansion at time m

Applying one more within-block token x_m after the switch gives four paths, corresponding to the four entries of R :

State at time m (first token after switch), with single normaliser Z :

$$\begin{aligned} \langle \psi(m) | = \frac{1}{Z} \bigg[& \underbrace{p \frac{1}{2}}_{\text{was } P_1, \text{ switched to } P_2} \langle 2 | \otimes \langle \eta_0^{(1)} | T_{x_1}^{(1)} \dots T_{x_{m-1}}^{(1)} \otimes \langle \eta_0^{(2)} | T_{x_m}^{(2)} \\ & + \underbrace{(1-p) \frac{1}{2}}_{\text{was } P_1, \text{ stayed in } P_1} \langle 1 | \otimes \langle \eta_0^{(1)} | T_{x_1}^{(1)} \dots T_{x_m}^{(1)} \otimes \langle \eta_0^{(2)} | \\ & + \underbrace{p \frac{1}{2}}_{\text{was } P_2, \text{ switched to } P_1} \langle 1 | \otimes \langle \eta_0^{(1)} | T_{x_m}^{(1)} \otimes \langle \eta_0^{(2)} | T_{x_1}^{(2)} \dots T_{x_{m-1}}^{(2)} \\ & + \underbrace{(1-p) \frac{1}{2}}_{\text{was } P_2, \text{ stayed in } P_2} \langle 2 | \otimes \langle \eta_0^{(1)} | \otimes \langle \eta_0^{(2)} | T_{x_1}^{(2)} \dots T_{x_m}^{(2)} \bigg], \end{aligned} \quad (13)$$

where the single global normaliser is the sum of all four unnormalised components:

$$Z = \frac{p}{2} (Z^{1 \rightarrow 2} + Z^{2 \rightarrow 1}) + \frac{1-p}{2} (Z^{1 \rightarrow 1} + Z^{2 \rightarrow 2}). \quad (14)$$

The four terms in (13) correspond to the four entries of the switch matrix R applied to the two initial regime components. After the switch, the two P_1 -regime terms (rows 2 and 3) can be combined into a single expression for the updated $\langle 1 |$ component, and similarly for P_2 . The resulting state is generically an entangled (non-product) state in $\mathbb{R}^2 \otimes \mathbb{R}^3 \otimes \mathbb{R}^3$.

5.3 Regime weight after the switch

The symmetric switch matrix R preserves the uniform regime weight:

$$\begin{bmatrix} \frac{1}{2} & \frac{1}{2} \end{bmatrix} R = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \end{bmatrix}.$$

However, the Bayesian regime weight λ_t returned by the observation history will in general *not* remain at $1/2$ (it is updated by the within-block observations via (6)), so λ after the switch is

$$\lambda_{\text{after}} = (1-p) \lambda_{\text{before}} + p(1 - \lambda_{\text{before}}). \quad (15)$$

Only if $\lambda_{\text{before}} = 1/2$ does $\lambda_{\text{after}} = 1/2$.

6 Full Update Cycle

One complete period of length m followed by a switch proceeds as:

1. **Initialise** with $\langle\psi(0)|$ as in (1).
2. **Within-block tokens** $x_1, \dots, x_m$: apply (6) at each step with global normaliser Z from (5).
3. **Switch boundary**: apply $S = R \otimes I_9$ as in (12), no normalisation needed.
4. Repeat from step 2 for the next block.

The full m -step propagator (before normalisation) is:

$$\langle\psi(m)| \propto \langle\psi(0)| (R \otimes I_9) U(x_m) \cdots U(x_1), \quad (16)$$

with a single global normaliser accumulated over all m steps.

7 Next-Token Predictive Distribution

7.1 The predictive probability as a contraction

The key observation is that the global normaliser Z computed at each within-block step *is* the next-token probability. This follows directly from the definition of $U(x)$ and total probability:

Next-token predictive probability at time t :

$$P(x_{t+1} = x \mid x_{1:t}) = \langle\psi(t)| U(x) \mathbf{1}_{18} = Z(x), \quad (17)$$

where $\mathbf{1}_{18} = \mathbf{1}_2 \otimes \mathbf{1}_3 \otimes \mathbf{1}_3$ is the all-ones vector in $\mathbb{R}^{18}$ and $Z(x)$ is the normaliser (5) evaluated for token x . The distribution over all three tokens is obtained by evaluating (17) for each $x \in \{A, B, C\}$. The three values sum to one by construction.

7.2 Regime-mixture decomposition

Expanding $U(x)$ using (2) and contracting with $\mathbf{1}_{18}$:

$$\begin{aligned} P(x_{t+1} = x \mid x_{1:t}) &= \langle\psi(t)| (|1\rangle\langle 1| \otimes T_x^{(1)} \otimes I_3 + |2\rangle\langle 2| \otimes I_3 \otimes T_x^{(2)}) \mathbf{1}_{18} \\ &= \underbrace{\lambda_t \langle\eta_t^{(1)}| T_x^{(1)} \mathbf{1}}_{P(x|r_t=1, x_{1:t}) \cdot \lambda_t} + \underbrace{(1 - \lambda_t) \langle\eta_t^{(2)}| T_x^{(2)} \mathbf{1}}_{P(x|r_t=2, x_{1:t}) \cdot (1 - \lambda_t)}. \end{aligned} \quad (18)$$

This is the **law of total probability over the regime**: the predictive distribution is a convex combination of the two per-regime predictive distributions, weighted by the current regime belief λ_t .

Define the per-regime token probability vectors (each in $\Delta_2 \subset \mathbb{R}^3$):

$$q_t^{(1)}(x) = \langle\eta_t^{(1)}| T_x^{(1)} \mathbf{1}, \quad q_t^{(2)}(x) = \langle\eta_t^{(2)}| T_x^{(2)} \mathbf{1}, \quad (19)$$

so that the predictive distribution over $\mathcal{X}$ is:

$$P(x_{t+1} \mid x_{1:t}) = \lambda_t q_t^{(1)}(x_{t+1}) + (1 - \lambda_t) q_t^{(2)}(x_{t+1}). \quad (20)$$

Remark 7.1 (Connection to the single-process MSP). For a single Mess3 process (i.e. $\lambda_t = 1$ always), $q_t^{(1)}(x) = \langle\eta_t^{(1)}| T_x^{(1)} \mathbf{1}$ is exactly the Riechers next-token probability. The switching process augments this with a second term governed by the regime uncertainty λ_t .

7.3 Obtaining the conditional belief vectors $\langle \eta_t^{(i)} |$

The conditional active-chain beliefs required in (18) are recovered from the full belief matrices $M_t^{(i)}$ (the $\mathbb{R}^{3 \times 3}$ blocks of $\langle \psi(t) |$):

$$\langle \eta_t^{(1)} | = \frac{\mathbf{1}_3^\top M_t^{(1)}}{\lambda_t} \in \Delta_2, \quad \langle \eta_t^{(2)} | = \frac{M_t^{(2)} \mathbf{1}_3}{1 - \lambda_t} \in \Delta_2, \quad (21)$$

where $\lambda_t = \mathbf{1}_3^\top M_t^{(1)} \mathbf{1}_3$ and $1 - \lambda_t = \mathbf{1}_3^\top M_t^{(2)} \mathbf{1}_3$.

7.4 Predictive distribution at the switch boundary

Immediately after the switch operator $S = R \otimes I_9$ is applied, the regime weights become $\lambda_{\text{after}} = (1-p)\lambda_{\text{before}} + p(1-\lambda_{\text{before}})$ and the conditional beliefs $\langle \eta_{\text{after}}^{(i)} |$ are the regime-mixed combinations from the four-path expansion (13). Concretely, combining the two terms that land in regime 1 after the switch:

$$\langle \eta_{\text{after}}^{(1)} | = \frac{(1-p)\lambda_{\text{before}}}{\lambda_{\text{after}}} \langle \eta_{\text{before}}^{(1)} | + \frac{p(1-\lambda_{\text{before}})}{\lambda_{\text{after}}} \langle \eta_{\text{before}}^{(2)} |, \quad (22)$$

and symmetrically for $\langle \eta_{\text{after}}^{(2)} |$. The next-token prediction at the first timestep of a new block then uses these blended conditional beliefs in (20).

7.5 Full predictive computation at each timestep

The recursive procedure at every timestep t is:

1. **Extract** λ_t and $\langle \eta_t^{(i)} |$ from the current belief state $\langle \psi(t) |$.
2. **Compute** the per-regime token probability vectors for each $x \in \{A, B, C\}$:

$$q_t^{(i)}(x) = \langle \eta_t^{(i)} | T_x^{(i)} \mathbf{1}, \quad i = 1, 2.$$

3. **Form** the predictive distribution:

$$P(x_{t+1} = x \mid x_{1:t}) = \lambda_t q_t^{(1)}(x) + (1 - \lambda_t) q_t^{(2)}(x), \quad \forall x \in \mathcal{X}.$$

4. **Observe** x_{t+1} , set $Z = P(x_{t+1} \mid x_{1:t})$, and **update**:

$$\langle \psi(t+1) | = \langle \psi(t) | \frac{U(x_{t+1})}{Z}.$$

5. If $t+1$ is a multiple of m , apply the switch: $\langle \psi | \leftarrow \langle \psi | (R \otimes I_9)$.

Remark 7.2 (Computational cost). Each step requires two matrix-vector products $\langle \eta_t^{(i)} | T_x^{(i)}$ (one per process, one per token in $\mathcal{X}$), giving $2 \times 3 = 6$ products of a 3-vector against a 3×3 matrix. The full belief state update costs one 18×18 block-diagonal matrix application. Maintaining the full $M_t^{(i)}$ matrices costs $2 \times 9 = 18$ floats rather than the 6 floats of the marginalised 6-dim representation — a modest overhead for exact Bayesian inference.

8 Converting Between $\langle \psi(t) |$ and $(M_t^{(1)}, M_t^{(2)})$

The 18-dimensional row vector $\langle \psi(t) |$ and the pair of 3×3 matrices $(M_t^{(1)}, M_t^{(2)})$ are *two representations of exactly the same object*. This section makes the bijection explicit in both directions, derives what the update operators $U(x)$ and S look like in each representation, and reconciles the two.

8.1 Index structure of $\langle\psi(t)|$

$\langle\psi(t)|$ lives in the tensor product $\mathbb{R}^2 \otimes \mathbb{R}^3 \otimes \mathbb{R}^3$. We order the 18 basis elements as (r, s_1, s_2) with $r \in \{1, 2\}$, $s_1 \in \{0, 1, 2\}$, $s_2 \in \{0, 1, 2\}$, varying s_2 fastest (row-major within each regime block):

$$\text{flat index} = (r - 1) \cdot 9 + s_1 \cdot 3 + s_2 \in \{0, \dots, 17\}.$$

The first nine entries ($r = 1$, indices 0–8) form the *regime-1 block*; the last nine ($r = 2$, indices 9–17) form the *regime-2 block*.

8.2 $\langle\psi(t)| \longrightarrow (M_t^{(1)}, M_t^{(2)})$: reshaping the vector into two matrices

Vector-to-matrix conversion: Read off the two 3×3 matrices by splitting the 18-vector at its midpoint and reshaping each half:

$$M_t^{(r)}[s_1, s_2] = \langle\psi(t)|[(r, s_1, s_2)] = \langle\psi(t)|[(r - 1) \cdot 9 + s_1 \cdot 3 + s_2]. \quad (23)$$

In words: the regime- r block of $\langle\psi(t)|$, taken as a length-9 sub-vector and reshaped row by row, gives $M_t^{(r)}$.

Written out explicitly for $r = 1$:

$$M_t^{(1)} = \begin{pmatrix} \psi_0 & \psi_1 & \psi_2 \\ \psi_3 & \psi_4 & \psi_5 \\ \psi_6 & \psi_7 & \psi_8 \end{pmatrix}, \quad M_t^{(2)} = \begin{pmatrix} \psi_9 & \psi_{10} & \psi_{11} \\ \psi_{12} & \psi_{13} & \psi_{14} \\ \psi_{15} & \psi_{16} & \psi_{17} \end{pmatrix},$$

where $\psi_k = \langle\psi(t)|[k]$. The row index of $M_t^{(r)}$ is always s_1 (state of P_1) and the column index is always s_2 (state of P_2).

8.3 $(M_t^{(1)}, M_t^{(2)}) \longrightarrow \langle\psi(t)|$: stacking the matrices into a vector

Matrix-to-vector conversion: Flatten each matrix in row-major order and concatenate:

$$\langle\psi(t)| = \left[\underbrace{M_t^{(1)}[0, 0], M_t^{(1)}[0, 1], M_t^{(1)}[0, 2], M_t^{(1)}[1, 0], \dots, M_t^{(1)}[2, 2]}_{\text{regime-1 block, entries 0-8}}, \underbrace{M_t^{(2)}[0, 0], \dots, M_t^{(2)}[2, 2]}_{\text{regime-2 block, entries 9-17}} \right]. \quad (24)$$

8.4 What the update operators look like in 18-dim vs. matrix form

Within-block update operator $U(x)$. In the 18-dimensional space the operator is (Section 4.1):

$$U(x) = |1\rangle\langle 1| \otimes T_1(x) \otimes I_3 + |2\rangle\langle 2| \otimes I_3 \otimes T_2(x).$$

This is block-diagonal with two 9×9 blocks:

$$U(x) = \begin{pmatrix} T_1(x) \otimes I_3 & 0 \\ 0 & I_3 \otimes T_2(x) \end{pmatrix}.$$

The update is $\langle\psi(t+1)| = \langle\psi(t)| U(x) / Z$.

To derive the matrix-level update we apply the row-major vectorisation identity: for matrices A, B, C of compatible sizes, $\text{rowvec}(ABC) = \text{rowvec}(B)(C \otimes A)$.

Regime-1 block. The regime-1 sub-vector of $\langle\psi(t)|$ is $\text{rowvec}(M_t^{(1)})$, and it right-multiplies the block $T_1(x) \otimes I_3$:

$$\text{rowvec}(M_{t+1}^{(1)}) \propto \text{rowvec}(M_t^{(1)}) (T_1(x) \otimes I_3).$$

Applying the identity with $A = T_1(x)^\top$, $B = M_t^{(1)}$, $C = I_3$:

$$M_{t+1}^{(1)} \propto T_1(x)^\top M_t^{(1)}. \quad (25)$$

Regime 1 active: $T_1(x)^\top$ **left-multiplies** $M_t^{(1)}$. This updates the s_1 (row, active P_1) dimension for every fixed column s_2 , exactly as the Riechers update $\langle\eta| \rightarrow \langle\eta| T_1(x)$ applied to each frozen- s_2 slice.

Regime-2 block. The regime-2 sub-vector right-multiplies $I_3 \otimes T_2(x)$. With $A = I_3$, $B = M_t^{(2)}$, $C = T_2(x)$:

$$M_{t+1}^{(2)} \propto M_t^{(2)} T_2(x). \quad (26)$$

Regime 2 active: $T_2(x)$ **right-multiplies** $M_t^{(2)}$. This updates the s_2 (column, active P_2) dimension for every fixed row s_1 .

Remark 8.1 (Asymmetry between the two updates). The two matrix updates are *not* symmetric: $M^{(1)}$ is updated by left-multiplication (transpose of $T_1(x)$) while $M^{(2)}$ is updated by right-multiplication ($T_2(x)$ directly). This asymmetry is a direct consequence of the layout choice that s_1 is always on rows and s_2 on columns: the active dimension is on *rows* for regime 1 but on *columns* for regime 2. Both are consistent Riechers-style updates; the difference is purely notational. If one prefers symmetric notation, one may instead store $\widetilde{M}_t^{(1)} = (M_t^{(1)})^\top$ (transpose), in which case the regime-1 update becomes $\widetilde{M}_{t+1}^{(1)} \propto \widetilde{M}_t^{(1)} T_1(x)$ — a right-multiply matching the regime-2 form.

The normaliser $Z = Z_1 + Z_2$ in matrix form is:

$$Z_1 = \mathbf{1}_3^\top (T_1(x)^\top M_t^{(1)}) \mathbf{1}_3 = \mathbf{1}_3^\top M_t^{(1)} T_1(x) \mathbf{1}_3, \quad Z_2 = \mathbf{1}_3^\top M_t^{(2)} T_2(x) \mathbf{1}_3. \quad (27)$$

Switch boundary operator $S = R \otimes I_9$. In 18-dim: $\langle\psi_{\text{after}}| = \langle\psi_{\text{before}}| (R \otimes I_9)$.

Since $R \otimes I_9$ permutes the two 9-entry regime blocks with weights $(1-p)$ and p (for the symmetric case), the matrix-level effect is straightforwardly:

$$M_{\text{after}}^{(1)} = (1-p) M_{\text{before}}^{(1)} + p M_{\text{before}}^{(2)}, \quad M_{\text{after}}^{(2)} = p M_{\text{before}}^{(1)} + (1-p) M_{\text{before}}^{(2)}. \quad (28)$$

This requires no reshaping: the two 3×3 matrices are mixed directly with scalar weights. No normalisation is needed since R is stochastic.

Summary: full correspondence.

Operation	18-dim form $\langle\psi \mapsto \langle\psi O$	Matrix form
Within-block (regime 1 active)	$\langle\psi (1\rangle\langle 1 \otimes T_1(x) \otimes I_3)$	$M^{(1)} \leftarrow T_1(x)^\top M^{(1)}$
Within-block (regime 2 active)	$\langle\psi (2\rangle\langle 2 \otimes I_3 \otimes T_2(x))$	$M^{(2)} \leftarrow M^{(2)} T_2(x)$
Normalise	divide by $Z = Z_1 + Z_2$	divide both matrices by Z
Switch boundary	$\langle\psi (R \otimes I_9)$	eqs. (28)

9 Reading Out the State of Each Individual Process

The full belief state $\langle\psi(t)|$, stored as two 3×3 matrices $(M_t^{(1)}, M_t^{(2)})$, contains more information than just the next-token prediction. This section explains exactly what you know about each process’s hidden Markov state $\sigma^{(i)}$ at every timestep t .

There are three levels of answer, each obtained by a different marginalisation of $(M_t^{(1)}, M_t^{(2)})$.

9.1 Level 1: Regime-conditional beliefs (four quantities)

The most informative statements condition on which regime is currently active. There are two processes and two possible active regimes, giving four regime-conditional beliefs, all recovered as row vectors in Δ_2 :

Four regime-conditional beliefs:

$$\text{Active } P_1 \text{ state : } \langle a_t^{(1)} | = \frac{M_t^{(1)} \mathbf{1}_3}{\lambda_t} \approx \text{belief over } \sigma^{(1)} \text{ given } r_t = 1, x_{1:t} \quad (29)$$

$$\text{Frozen } P_2 \text{ state (given } r_t = 1) : \langle f_t^{(2|1)} | = \frac{\mathbf{1}_3^\top M_t^{(1)}}{\lambda_t} \approx \text{belief over } \sigma^{(2)} \text{ given } r_t = 1, x_{1:t} \quad (30)$$

$$\text{Frozen } P_1 \text{ state (given } r_t = 2) : \langle f_t^{(1|2)} | = \frac{M_t^{(2)} \mathbf{1}_3}{1 - \lambda_t} \approx \text{belief over } \sigma^{(1)} \text{ given } r_t = 2, x_{1:t} \quad (31)$$

$$\text{Active } P_2 \text{ state : } \langle a_t^{(2)} | = \frac{\mathbf{1}_3^\top M_t^{(2)}}{1 - \lambda_t} \approx \text{belief over } \sigma^{(2)} \text{ given } r_t = 2, x_{1:t} \quad (32)$$

The key asymmetry is:

- The **active** belief ($\langle a_t^{(i)} |$) is Bayesian-updated at every step within the current block — it carries information from every token seen since the last switch.
- The **frozen** belief ($\langle f_t^{(j|i)} |$ for $j \neq i$) is the distribution over the inactive chain’s state. Under variant [2a] this is preserved exactly from the moment P_j became inactive. It has not been updated by any token in the current block.

Remark 9.1 (Why there are four beliefs, not two). A naive 6-dim representation stores one belief per process. But the correct picture requires *two* beliefs per process: an active belief and a frozen belief for each of the two possible active regimes. The four quantities above are contained

in the off-diagonal structure of the 3×3 matrices $M_t^{(i)}$ — they are not recoverable from the marginalised 6-dim state.

9.2 Level 2: Marginal beliefs over each process (two quantities)

If you want to know the state of process P_i *regardless of which regime is currently active*, marginalise over the regime:

Unconditional marginal beliefs over each process state:

$$P(\sigma_t^{(1)} = s_1 \mid x_{1:t}) = [(M_t^{(1)} + M_t^{(2)}) \mathbf{1}_3]_{s_1}, \quad (33)$$

$$P(\sigma_t^{(2)} = s_2 \mid x_{1:t}) = [\mathbf{1}_3^\top (M_t^{(1)} + M_t^{(2)})]_{s_2}. \quad (34)$$

Compactly, defining $M_t = M_t^{(1)} + M_t^{(2)}$:

$$\langle \mu_t^{(1)} \mid = M_t \mathbf{1}_3 \in \Delta_2, \quad \langle \mu_t^{(2)} \mid = \mathbf{1}_3^\top M_t \in \Delta_2.$$

These marginals can be written as regime-weighted mixtures of the four level-1 quantities:

$$\langle \mu_t^{(1)} \mid = \lambda_t \langle a_t^{(1)} \mid + (1 - \lambda_t) \langle f_t^{(1|2)} \mid, \quad (35)$$

$$\langle \mu_t^{(2)} \mid = \lambda_t \langle f_t^{(2|1)} \mid + (1 - \lambda_t) \langle a_t^{(2)} \mid. \quad (36)$$

Equation (35) says: your overall belief about P_1 's state is a mixture of “ P_1 is active and has belief $\langle a_t^{(1)} \mid$ ” (weight λ_t) and “ P_1 is frozen and has belief $\langle f_t^{(1|2)} \mid$ ” (weight $1 - \lambda_t$). A sharp active belief and a sharp frozen belief do not in general agree, so $\langle \mu_t^{(1)} \mid$ will be more diffuse than either alone.

9.3 Level 3: Joint distribution over both chain states

The joint distribution over $(\sigma_t^{(1)}, \sigma_t^{(2)})$, marginalising only the regime r_t , is:

$$P(\sigma_t^{(1)} = s_1, \sigma_t^{(2)} = s_2 \mid x_{1:t}) = M_t^{(1)}[s_1, s_2] + M_t^{(2)}[s_1, s_2] = M_t[s_1, s_2]. \quad (37)$$

Remark 9.2 (Why M_t alone is insufficient as a state). $M_t = M_t^{(1)} + M_t^{(2)}$ is the 9-dimensional quantity that the earlier (incorrect) notes were tracking. It correctly represents the joint distribution over both chain states (37). However, it is *not sufficient* to propagate the belief forward in time: to compute the next token probability or perform the switch update you need λ_t and the regime-conditional beliefs, which require knowing $M_t^{(1)}$ and $M_t^{(2)}$ *separately*, not just their sum.

9.4 Interpretation: what the frozen memory gives you

The four level-1 beliefs have qualitatively different characters depending on the position within a block:

Belief	Updated by	Quality
$\langle a_t^{(i)} \mid$ (active)	Every token in current block	Sharpens with block length
$\langle f_t^{(j i)} \mid$ (frozen, $j \neq i$)	No tokens since last switch	Preserved from end of last P_j block

At the start of a fresh block (just after a switch), the newly active process begins from its frozen belief $\langle f_{\text{switch}}^{(i)} |$, which encodes exactly what the observer knew about P_i 's state the last time it was active. As tokens accumulate, $\langle a_t^{(i)} |$ moves away from this starting point and becomes more informative.

Concretely, after k tokens in the current block under process P_i , the active belief satisfies:

$$\langle a_t^{(i)} | \propto \langle f_{\text{switch}}^{(i)} | T_{x_{t-k+1}}^{(i)} \cdots T_{x_t}^{(i)}, \quad (38)$$

which is exactly the single-process Riechers update run forward from the frozen starting point $\langle f_{\text{switch}}^{(i)} |$. This confirms that the frozen memory acts as a *warm start*: the active process does not need to re-synchronise from $\pi^{(i)}$ at each block — it resumes from wherever inference had reached at the end of its previous active period.

10 Comparison of Variants

	[1] Restart	[2a] Frozen	[2b] Latent
Inactive chain	Resets to $\pi^{(i)}$ on switch	Frozen (identity)	Evolves under $T^{(i)}$
Within-block update	$ 1\rangle\langle 1 \otimes T_x^{(1)} \otimes I$ + $ 2\rangle\langle 2 \otimes I \otimes T_x^{(2)}$	Same	$ 1\rangle\langle 1 \otimes T_x^{(1)} \otimes T^{(2)}$ + $ 2\rangle\langle 2 \otimes T^{(1)} \otimes T_x^{(2)}$
Switch operator	$R \otimes I_9$, then reset inactive	$R \otimes I_9$	$R \otimes I_9$
Belief dimension	$1 + 2 + 2 = 5$ (regime, two active η)	17 (two 3×3 matrices)	17
Memory of inactive	None	Exact, indefinite	Decays as $T^t \rightarrow \pi$

References

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